

Bengali Fake Review Detection – Technical Overview

This project focuses on detecting fake Bengali product reviews using modern NLP techniques. We explored multiple transformer-based models and large language models (LLMs) to assess their effectiveness in classifying deceptive reviews in Bengali text. The work includes dataset preprocessing, model experimentation, and ensemble strategies to improve classification reliability. Source code and detailed experimental results are not included here as the work is still being refined.

1. Dataset Preparation

We used a publicly available Bengali fake review dataset. Several preprocessing steps were applied to improve data quality and model performance, including:

- Duplicate review removal and label consistency checking
- Text cleaning and normalization
- Data augmentation for the minority class
- Balanced dataset creation (approximately 1:1 real vs fake)
- Stratified train/validation/test split for reliable evaluation

These steps helped ensure consistent training conditions and fair model comparison.

2. Transformer-Based Models

Multiple transformer-based NLP models were fine-tuned for Bengali fake review classification. These models were evaluated comparatively to understand performance differences between multilingual and Bengali-specialized architectures.

Models explored:

- BanglaBERT
- XLM-RoBERTa (XLM-R)
- mBERT (Multilingual BERT)
- BanglaT5
- Gemma-2B

Overall, Bengali-specialized transformer models generally showed more stable performance due to stronger language-specific representation.

3. Large Language Model (LLM) Experiments

We explored several large language model (LLM) approaches for Bengali fake review detection using supervised fine-tuning, prompt-based training, zero/few-shot evaluation, commercial model testing, and ensemble techniques. The goal was to understand how different LLM architectures handle Bengali text classification and how fine-tuning strategies influence performance.

Supervised / Classification-Based LLM Fine-Tuning

For supervised classification experiments, multiple Bengali and multilingual LLM checkpoints were fine-tuned by adding a classification head. Some models were fine-tuned using parameter-efficient approaches such as LoRA, while others were directly fine-tuned for binary classification after dataset preprocessing and balancing. During inference, predictions were obtained directly from the classifier output layer.

Models used in this stage:

- Bangla-s1k-llama-3.2-3B (Classification head, LoRA fine-tuned)
- culturax-base-3b-seqcls (Classification head fine-tuned)
- BanglaLLama-3.2-3b-alpaca (Classification head fine-tuned)
- Bangla-s1k-qwen-2.5-3B (Classification head, LoRA fine-tuned)

Additional large-scale experiment:

- Bangla-LLaMA-7B-Instruct (Classification head, LoRA fine-tuned)
- Bangla-LLaMA-7B-Instruct (Full precision FP32 fine-tuning – ongoing)

These experiments helped evaluate how supervised fine-tuning improves classification reliability for Bengali fake review detection.

Prompt-Finetuned Causal Language Models

Prompt-formatted fine-tuning experiments were also conducted using causal language models. In this setup, the classification problem was converted into a text-generation task where models generated labels such as “fake” or “real” based on structured prompts. Predictions were mapped from generated outputs to classification labels.

Models explored:

- Bangla-s1k-qwen-2.5-3B (Prompt-finetuned causal LM)
- Bangla-s1k-llama-3.2-3B (Prompt-finetuned causal LM)
- BanglaLLama-3.2-3b-alpaca (Prompt-finetuned causal LM)
- culturax-base-3b (Prompt-finetuned causal LM)

This allowed comparison between classification-head fine-tuning and generative prompt-based classification.

Zero-shot and Few-shot LLM Evaluation

Several pretrained LLMs were evaluated without task-specific training using zero-shot and few-shot prompting strategies. Prompt variations were tested to observe sensitivity to prompt design and contextual examples.

Models tested:

- Bangla-s1k-qwen-2.5-3B (Zero-shot and Few-shot)
- BanglaLLM prompt variants (Zero-shot and Few-shot with prompt modification)
- BanglaLLama-3.2-1b-bangla-alpaca-orca-instruct-v0.0.1 (Zero-shot and Few-shot)
- BanglaLLama-3.2-3b-bangla-alpaca-orca-instruct-v0.0.1 (Zero-shot and Few-shot)
- bangla-llama-7b-instruct-v0.1 (Zero-shot and Few-shot)

These experiments helped assess baseline generalization capability without additional training.

Commercial LLM Evaluation

To compare open-source models with proprietary systems, selected samples from the dataset were manually tested using commercial LLM APIs without additional fine-tuning. Testing involved manually designed prompts applied to representative review samples.

Commercial models tested:

- Gemini
- ChatGPT
- DeepSeek

This comparison helped evaluate practical applicability of commercial LLMs for Bengali fake review detection.

Ensemble Techniques

To improve classification robustness, predictions from multiple fine-tuned models were combined using several ensemble strategies:

- Majority voting
- Soft voting
- Weighted soft voting
- Stacking ensemble

These ensemble approaches were applied at inference time to stabilize predictions across models.

Evaluation Approach

Model performance was evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score. Additional ongoing analysis includes robustness evaluation, calibration assessment, and comparative statistical analysis.

Detailed experimental results, training configurations, and implementation specifics are intentionally not included here as the work is still being refined.

6. Notes

This document provides a conceptual and methodological overview only. Detailed training configurations, implementation code, and complete experimental results are not included here as the project is still being refined.